

August 3, 2026

Biobank Working Group



Agenda

- 12:00–12:10 Updates
 - Letter of Intent to AZA Board
 - IMLS grant application (Nov 13)
 - AZA MSAG Webinar (Oct 20)
- 12:10–1:25 Objectives
- 1:25–1:30 Wrap-up



Issues

Vision

Goals

Objectives

Actions

By 2055, biological specimens and associated data from biodiversity institutions are mobilized seamlessly at scale—driving discovery, strengthening resilient populations, and safeguarding species through fully integrated *ex situ* and *in situ* conservation and research.

Goal 1: By 2032, biological samples and associated data from across participating zoological and aquarium institutions are seamlessly **discoverable** and **accessible** for the global advancement of non-commercial research and conservation.

Goal 2: By 2032, participating zoological and aquarium institutions **strategically** and **systematically** steward biological samples and associated data through contributions to **standardized** biobanking, maximizing benefit for global biodiversity knowledge and conservation.

Brainstorming objectives

Biobank Alliance Working Group, Phase 2, 2026

Objectives - July Homework

Please add draft objectives to this document by **Friday July 24**. Add objectives under the appropriate goal and include your name to help facilitate discussion. Objectives will be synthesized and refined in breakout groups.

Objectives describe the major accomplishments/milestones **required** to achieve the goal. A strong objective:

- Describes **what must be accomplished** to achieve the goal.
- Describes a measurable outcome or milestone.
- Is more specific than a goal, but broader than individual actions.
- Includes a realistic, appropriate timeframe.
- Is ambitious but realistic within the scope of the Working Group & participating institutions.

Don't worry about writing the perfect objective. We're looking for raw material for your breakout groups to synthesize and refine together. It's fine if objectives overlap or build on one another. Similar ideas can reveal areas of convergence. Feel free to add comments on, or expand, existing ideas in addition to adding your own.

Goal 1: By 2032, biological samples and associated data from across participating zoological and aquarium institutions are seamlessly *discoverable* and *accessible* for the global advancement of non-commercial research and conservation.

- (2 yr): By 2029, 10 institutions have both a designated, public biobank contact and ≥ 50 samples publicly discoverable. -Rachel Johnston (example)
- (5 yr): By 2032, participating biobanking organizations share their sample inventories to identify collection gaps and establish strategic priorities for future sample acquisition. - Cynthia Steiner
- (1 yr): By 2028, 10 institutions have bought-in and designated a facility contact for coordination - John Griffioen
- (2 yr): By 2029, the committed institutions have begun inputting existing samples into a coordinated database and begun beta testing access - John Griffioen
- (3 yr): By 2030, 5-10 additional institutions have been recruited and beta testing with access has begun with select research partners - John Griffioen
- (4 yr): By 2031, a beta version of a searchable database is available to participating zoological and aquarium institutions and there are 20 committed partners - John Griffioen
- (5 yr): By 2032, 25 partner institutions are routinely adding available samples to a searchable database - John Griffioen
- By 2028, partner with the Ape TAG and great ape SSPs to use as a pilot case for developing multiinstitutional inventory, sample and data sharing, research permissions, etc. (Ape species would be an attractive pilot case given species holdings of the current 10 working group institutions and current active research projects.) - Eric Baitchman

Goal 2: By 2032, participating zoological and aquarium institutions strategically and systematically steward biological samples and associated data through contributions to *standardized* biobanking, maximizing benefit for global biodiversity knowledge and conservation.

- (5 yr): By 2032, participating institutions adopt a shared minimum metadata standard that enables biospecimens & associated data to be searched and interpreted consistently across institutions. -Rachel Johnston (example)
- (2 yr): By 2029, participating institutions seamlessly collaborate with SSP coordinators and programs to identify and collect priority samples for biobanking in support of population management. - Cynthia Steiner
- (2 yr?): I would like to see that we successfully persuade the AZA to elevate biobanking as a strategic priority for institutions, encouraging their leadership and boards to explicitly define and communicate their institution's stance on biobanking and level of commitment. As levels of commitment will be influenced by financial resources, infrastructure, and personnel capacity, institutions will have different biological

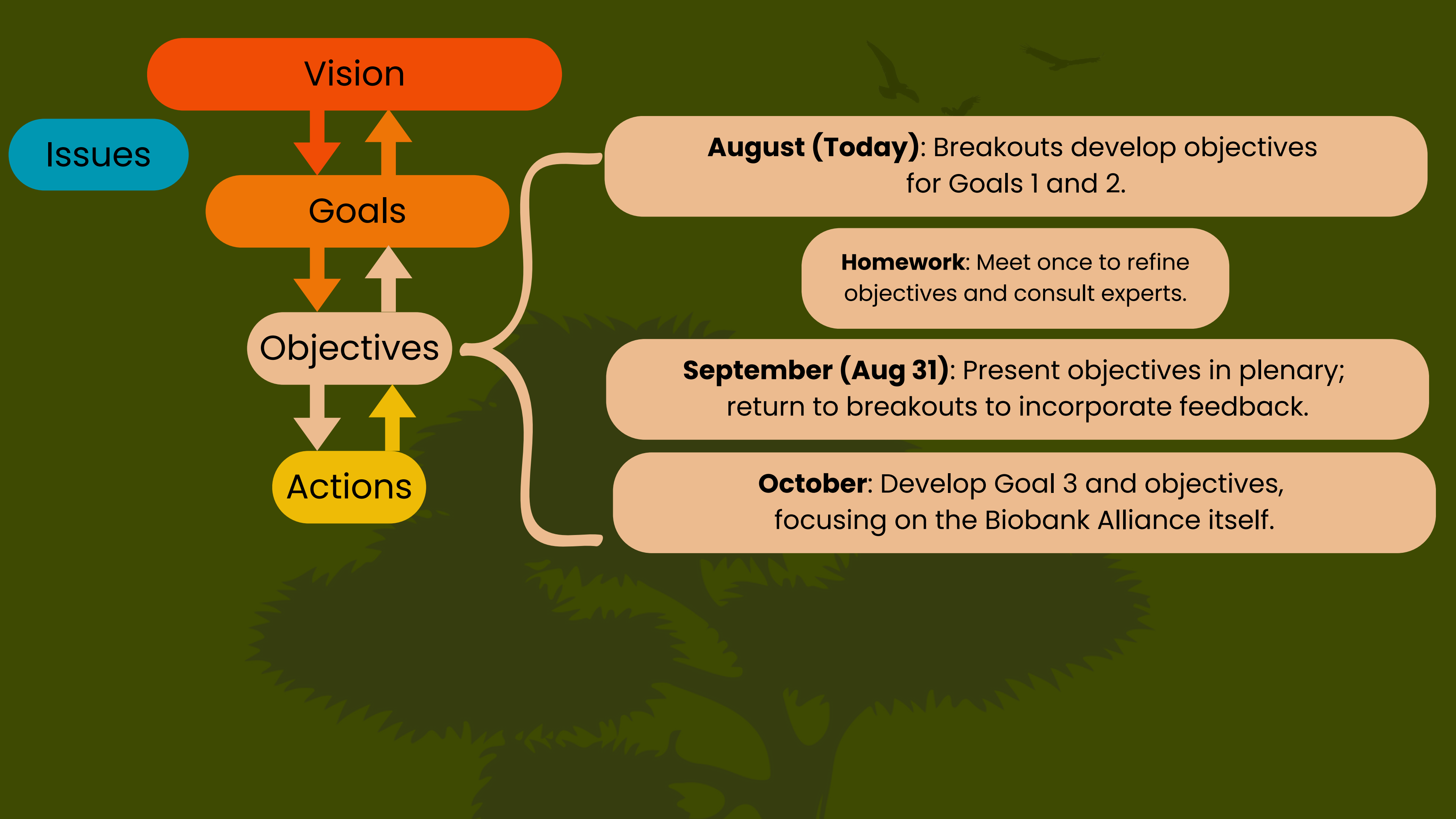
Biobank Alliance Working Group, Phase 2, 2026

- sampling strategies. This may seem obvious, but in the end biobanking depends on how individual employees are expected to spend their time. If they don't know the institutional level of commitment, standardizing practices won't happen because of the effort to get everyone on the same page. - Cynthia Frasier
- (1 yr): By 2028, biobanking goals (of AZA and participating institutions) are identified. Shared goals that most benefit biodiversity and conservation research are then agreed upon. - Erin Ehmke
 - (2 yr): By 2029, mechanics of shared goals are decided: prioritization of species, sample types, sample volume, and preservation methods are identified and communicated to participating institutions. - Erin Ehmke
 - (timeline?): Agreements in place for long-term storage of prioritized specimens as well as who can utilize them, and who approves their use for which projects. - Erin Ehmke
 - Develop a tiered Biobank Alliance participation model, with access to training and protocols, to make standardization an achievable goal for small and large institutions. - Cynthia Frasier

Goal 3: Alliance Governance & Sustainability (Looming; for October Discussion)

As you draft objectives, you may identify ideas that are essential to the long-term success of the Biobank Alliance itself but extend beyond institutional biobanking practices, such as governance and sustainable funding. Please capture those ideas here. These ideas will be considered and discussed as we continue refining the action plan.

- (2 yr): By 2029, sustainable, long-term funding sources (federal, regional, and private) are identified to support national biobanking programs and secure strategic biobanking priorities. - Cynthia Steiner



Objectives : the major **accomplishments**/milestones **required** to achieve the goal.

A strong objective:

- Describes a **major accomplishment**, not an activity.
- Is **required** to achieve the goal.
- Directly addresses the **priority issue**.
- Leaves room for many possible actions.
- Is within the influence of the Working Group and their participating institutions.



Goal for today

Identify a concise set of well-defined, non-overlapping **essential** objectives that this Working Group is prepared to champion in order to achieve the shared goal.



Refinement

For every objective ask:

- Is this a **required accomplishment** to achieve the goal?
- Can it be combined with another objective, or is anything important missing?
- If we could only keep 3–6 objectives, does this make the cut?
- Could this Working Group realistically lead or catalyze this accomplishment?
- Are we prepared to invest the time needed to make it happen?
- If this became one of our objectives, who in this room would help make it happen?



Each breakout group will:

- Have a facilitator (Danica or Rachel)
- Need a recorder and a reporter.
- Brainstorm objectives, then prioritize and refine
- Be responsible for bringing forward a thoughtful, well-supported recommendation of objectives for the full Working Group to consider.
- Meet at least once before next meeting to
 - refine your proposed objectives
 - consult additional experts as needed



Advisory Network

NAME	ORGANIZATION REPRESENTING	OTHER AFFILIATIONS	EMAIL
ADVISORY NETWORK			
Megan Brown	AZA		mbrown@aza.org
Shelly Grow	AZA		sgrow@aza.org
Budhan Pukazhenthi	AZA Biobanking SAG	Smithsonian's National Zoo and Conservation Biology Institute	pukazhenthib@si.edu
Andrea Putnam	AZA Molecular SAG	Dallas Zoo	asputnam@gmail.com
Jen Tobey	AZA Research and Technology Committee	San Diego Zoo Wildlife Alliance	jtobey@sdzwa.org
Terri Roth	Center for Conservation & Research of Endangered Wildlife	Cincinatti Zoo and Botanical Garden	Terri.Roth@cincinnati-zoo.org
Rachel Johnston	Center for Zoonomics	Zoo New England	rjohnston@zoonewengland.org
Ania Brown	EAZA		ania.brown@eaza.net
Marietta Danforth	Great Ape Heart Project	Detroit Zoological Society	gahpinfo@gmail.com
Andrew Torrance	Individual expertise	University of Kansas School of Law/MIT Sloan School of Management	torrance@mit.edu
Andy Richardson	Individual expertise	San Diego Zoo Wildlife Alliance	arichardson@sdzwa.org
Asako Chaille	Individual expertise	San Diego Zoo Wildlife Alliance; Molecular SAG	achaille@sdzwa.org
Carla Bonetti Madelaire	Individual expertise	San Diego Zoo Wildlife Alliance	cbonettimadelaire@sdzwa.org
Cynthia Steiner	Individual expertise	San Diego Zoo Wildlife Alliance	csteiner@sdzwa.org
Danica Wolfe	Individual expertise	Zoo New England; Center for Zoonomics	dwolfe@zoonewengland.org
Frank Oliaro	Individual expertise	John G. Shedd Aquarium	foliaro@sheddaquarium.org
Kameron Roth	Individual expertise	San Diego Zoo Wildlife Alliance	kroth@sdzwa.org
Kim Labuschagne	Individual expertise	SANBI	K.Labuschagne@sanbi.org.za
Michelle White	Individual expertise	Broad Institute of MIT & Harvard/Translational Medical Insights	mwhite@broadinstitute.org
Paula Mackie	Individual expertise	Toronto Zoo	pmackie@torontozoo.ca
Tahlia Perry	Individual expertise	Zoo New England	tperry@zoonewengland.org
Tanya Salvey	Individual expertise	John G. Shedd Aquarium	tsalvey@sheddaquarium.org
Oliver Ryder	IUCN SSC ABC Specialist Group	San Diego Zoo Wildlife Alliance	ORyder@sdzwa.org
Lauren Howard	Peel Therapeutics		lhoward@peeltx.com
Alex Garretson	Population Management Center (AZA)	Lincoln Park Zoo	agarretson@lpzoo.org
Ben Novak	Revive and Restore		ben@reviverestore.org
Rachel Thompson	Species360		rachel.thompson@species360.org
Greg Watkins-Colwell	ZooMu	Yale Peabody Museum	gregory.watkins-colwell@yale.edu
Elinor Karlsson	Individual expertise	Broad Institute of MIT & Harvard/Translational Medical Insights	elinor@broadinstitute.org
Lisa Moses	Individual expertise	Harvard Medical School	lisa_moses@hms.harvard.edu

Objectives - Goal 1

Recorder name:
Reporter name:
Breakout members:

Purpose

Generate a concise (3-6) set of well-defined, non-overlapping objectives that are essential to achieving the goal. We are not yet identifying specific actions.

Issue = the problem in the way of the vision.

Goal = the future condition we want to achieve.

Objectives = the major accomplishments required to achieve the goal.

Actions = the specific activities undertaken to accomplish the objectives (developed later).

Issue 1: Limited *discoverability* of available samples, associated data, and institutional points of contact (post-sampling stage)

Goal 1: By 2032, biological samples and associated data from across participating zoological and aquarium institutions are seamlessly *discoverable* and *accessible* for the global advancement of non-commercial research and conservation.

Step 1. Confirm the Goal

Review the priority issue and goal. As a group, discuss:

- Is this goal appropriately focused on addressing the primary priority issue, or is it trying to solve multiple issues equally?
- Keep in mind that a well-designed goal may also help address other issues, but it should remain anchored by one primary priority issue.
- If the goal is appropriately focused, proceed to developing objectives.
- If the goal appears to be trying to solve multiple issues equally, spend up to 10 minutes revising it so it has a clear primary focus before moving on to Step 2.

Sampling stage	Issue	Severity	Leverage	Feasibility	Sum
Post	Limited discoverability of available samples, associated data, and institutional points of contact	28	29	22	79
During	Lack of standardized practices, training, and knowledge transfer for biobank management	23	25	27	75
Pre	Unclear governance and responsibility for coordinated biobanking	26	25	20	71
Pre	Concerns about ownership, control, and future use of samples and data	26	26	19	71
Post	Lack of standardized metadata across institutions and biobanking systems	26	22	21	69
Pre	Lack of shared priorities, capabilities, and alignment across institutions for sample use	25	25	19	69
During	Lack of guidance and support for institutional biobanking	21	21	24	66
Post	Redundant institutional approval processes for sample and data access	23	23	19	65

Issues

Vision

Goals

Objectives

Actions

By 2055, biological specimens and associated data from biodiversity institutions are mobilized seamlessly at scale—driving discovery, strengthening resilient populations, and safeguarding species through fully integrated *ex situ* and *in situ* conservation and research.

Goal 1: By 2032, biological samples and associated data from across participating zoological and aquarium institutions are seamlessly **discoverable** and **accessible** for the global advancement of non-commercial research and conservation.

Goal 2: By 2032, participating zoological and aquarium institutions **strategically** and **systematically** steward biological samples and associated data through contributions to **standardized** biobanking, maximizing benefit for global biodiversity knowledge and conservation.

Wrap-up

- **Homework**

- Meet as a breakout group before 8/31 to complete your objectives and consult experts so you're ready for presentation.

- Next meeting: Monday August 31 (scheduled to avoid Labor Day)

